

The Tangent Vector

Given a vector function $\vec{r}(t)$, and the points $A = \vec{r}(a)$, $B = \vec{r}(a - h)$, and $C = \vec{r}(a + h)$. In the two-dimensional case, we have shown in [TM1](#) that the tangent line to $\vec{r}$ at A is perpendicular to $\overrightarrow{AS}$, where S is the center of the circumscribed circle of $\triangle ABC$ as $h \rightarrow 0$. This circle is called the osculating circle.

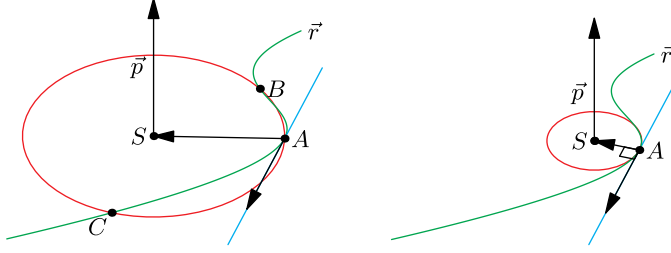


Figure 1: The blue line is the tangent line of $\vec{r}$ at A .

We will now examine a tangent line to a curve in space. We set $\vec{r}(0) = (0, 0, 0)$. Since the shape of the curve is what matters here, the results we derive will apply to all $\vec{r}(a)$, provided certain criteria are met. Therefore, we will write $\vec{r}'(0)$ and $\vec{r}''(0)$ as $\vec{r}'$ and $\vec{r}''$, respectively.

Center of the Circumscribed Circle

Let $\vec{u} = \vec{r}(-h)$ and $\vec{v} = \vec{r}(h)$. Then, $A = \vec{r}(0)$, $B = \vec{u}$, and $C = \vec{v}$. The circumscribed circle of $\triangle ABC$ lies in the plane spanned by $\vec{u}$ and $\vec{v}$, and thus has a normal vector $\vec{u} \times \vec{v}$. The center of this circle lies on the perpendicular bisector of the point $\frac{1}{2}\vec{u}$. This perpendicular bisector $l_1(t)$ can be expressed as:

$$l_1(t) = \frac{1}{2}\vec{u} + t\vec{u} \times (\vec{u} \times \vec{v})$$

Similarly, the perpendicular bisector $l_2(q)$ through the point $\frac{1}{2}\vec{v}$ can be expressed as:

$$l_2(q) = \frac{1}{2}\vec{v} + q\vec{v} \times (\vec{u} \times \vec{v})$$

These lines intersect at the center of the circumscribed circle, so at that point:

$$\frac{1}{2}\vec{u} + t\vec{u} \times (\vec{u} \times \vec{v}) = \frac{1}{2}\vec{v} + q\vec{v} \times (\vec{u} \times \vec{v})$$

By taking the dot product of $\vec{v}$ on both sides of this equation, we arrive at an expression for t :

$$t = \frac{|\vec{v}|^2 - \vec{u} \cdot \vec{v}}{2(\vec{u} \cdot \vec{v})^2 - |\vec{u}|^2|\vec{v}|^2}$$

Substituting this expression for t into the expression for $l_1(t)$, we find that:

$$\overrightarrow{AS} = \frac{1}{2}\vec{u} + \frac{|\vec{v}|^2 - \vec{u} \cdot \vec{v}}{2(\vec{u} \cdot \vec{v}) - |\vec{u}|^2|\vec{v}|^2}\vec{u} \times (\vec{u} \times \vec{v})$$

Furthermore, it can be shown¹ that:

$$\lim_{h \rightarrow 0} \overrightarrow{AS} = k(\vec{r}''|\vec{r}'| - \vec{r}'(\vec{r}' \cdot \vec{r}''))$$

where k is a constant. In this case, S is the center of the osculating circle of $\vec{r}$ at point A .

The Plane of the Circumscribed Circle

The circumscribed circle of $\triangle ABC$ must lie in the same plane as $\triangle ABC$, i.e., in the plane spanned by $\vec{u}$ and $\vec{v}$. Thus, $\vec{u} \times \vec{v}$ is a normal vector to this plane, and so $\vec{p} = \frac{1}{h^3}(\vec{u} \times \vec{v})$ is also a normal vector to the plane. It can be shown² that:

$$\lim_{h \rightarrow 0} \frac{1}{h^3}(\vec{u} \times \vec{v}) = \vec{r}' \times \vec{r}''$$

Geometric Interpretation of the Tangent Line

It is trivial to show that as $h \rightarrow 0$, $\overrightarrow{AS} \cdot \vec{r}' = 0$ and $\vec{p} \cdot \vec{r}' = 0$. Therefore, at point A , $\vec{r}'$ is perpendicular to both the vector pointing toward the center of the osculating circle and the normal vector of the plane in which the osculating circle lies.

Assumptions

We have made some assumptions:

- In the expression for t , we assumed that $\vec{u}$ and $\vec{v}$ are not parallel.
- In the expression for $\lim_{h \rightarrow 0} \overrightarrow{AS}$, k is only a constant if $\vec{r}'$ and $\vec{r}''$ exist and are non-zero.

¹E.g., using the *Taylor series* of a vector function

²Again, e.g., using a *Taylor series*